PAPER_951: Cooper Pair Phonon Coupling

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** bcs_superconductivity_uqff.py (CooperPairPhononCoupling) **Calculator:** CooperPairPhononCouplingCalc (CP4 #535) **CVW:** v2.0.0 compliant

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Abstract

We derive the effective Cooper-pair coupling strength via SCm phonon exchange at 1.25 THz. The coupling $V_{\text{eff}}(\omega, \Gamma) = V_{\text{SCm}} \cdot \Phi_{1.25\,\text{THz}}(\omega, \Gamma)$ is modulated by the phonon resonance profile $\Phi = \exp(-(\omega - \omega_{\text{SCm}})^2/(2\Gamma^2)) \cdot S_{26}$, yielding pair binding energy $E_{\text{pair}} = 2\Delta(T)$.

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1. Coupling Strength

$$V_{\text{eff}}(\omega, \Gamma) = V_{\text{SCm}} \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

2. Pair Binding Energy

$$E_{\text{pair}} = 2\Delta(T)$$

At on-resonance ($\omega = \omega_{\text{SCm}}$), $\Phi = S_{26}$ and coupling is maximized.

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3. Source Data

- **File:** bcs_superconductivity_uqff.py
- **Session:** 214
- **CP4 Class:** CooperPairPhononCouplingCalc (#535)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Cooper, L.N. (1956) – Phys. Rev. 104, 1189
3. PAPER_949 — BCS Gap Equation
4. PAPER_950 — BCS Critical Temperature
5. PAPER_955 — BCS Phonon Resonance
6. PAPER_957 — Cooper Pair Lagrangian

Cross-Links

Related Paper	Relationship
PAPER_949	Gap equation driven by this coupling
PAPER_950	T_c depends on V_{eff}
PAPER_952	Spectral ladder channels for pairing

Related Paper	Relationship
PAPER_955	Resonance Q-factor at ω_{SCm}

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	$5.0 \times 10^{-4}\,\text{day}^{-1}$	—	Magnetar spin-down
$[SSq]$	0.57	—	BH dynamics
β_i	0.603	—	Multi-system
ω_{SCm}	$2\pi \times 1.25$ THz	—	Phonon resonance
F_{UBi}/F_U	0.6	—	Buoyancy-gravity ratio

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Cooper pair binding	$E_{\text{pair}} = 2\Delta$ via $\Phi_{1.25\text{THz}}$	Mapped
Phonon mediation	$V_{\text{eff}} = V_{\text{SCm}} \cdot \Phi(\omega, \Gamma)$	Novel

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Cooper Pair Binding (BCS-SCm Phonon Mediation)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{pair}} = \bar{\psi}(i\partial\!\!\!/ - m)\psi + V_{\text{SCm}}(\bar{\psi}\psi)^2 \cdot \Phi_{1.25\text{THz}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{(i\partial\!\!\!/ - m)\psi = -2V_{\text{SCm}}\Phi \cdot (\bar{\psi}\psi)\psi}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ $\Phi_{1.25\text{THz}}$ phonon $\rightarrow$ Cooper pair binding $\rightarrow$ BCS condensate

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Cooper pair density scales with $\rho_{\text{SCm}} \cdot S_{26}$.

§B.2 DVP

Prime $p = 2$ encodes pair symmetry.

§B.3 BSH

$\mathrm{BSH}_{\mathrm{pair}} = \tanh(\beta_i \cdot \Delta / E_0) \cdot S_{26}$

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
$[SSq]$	0.57	Confirmed